Gate 0c — SPICE verification (KiCad page-2)

Energize: No. **Prior:** Gate 0b operator schematic verify PASS (`gate-0-sch-verify.md`).

Hard gate **before** redrawing page-2 as sourced **2.2 MΩ / 3.3 MΩ** networks and before Gate A stuffing.

Gate 0c cannot PASS without operator verification of the KiCad result. A green `run_bringup_sims.sh` exit is **mech prep only** — not gate closure.

Source of truth

Item	Path
Schematic	`bringup-supply.kicad_sch` rev `0.4-kicad-spice`
Export	`kicad-cli sch export netlist --format spice`
Mech runner	`../../sim/spice/run_bringup_sims.sh`
Last export	`../../sim/spice/kicad-out/bringup-supply.cir`

Hand decks `type_a_divider.cir` / `anode_pulse.cir` are reference only.

What is on the sheet (sim)

Ref	Role
Vhv	VDC −1000 V on **K** (sim-only)
R10–R18, Rbleed	Type A string + bleed (hardware)
Ra, Ca	Anode load 1.1 MΩ + 15 pF (sim-only)
Iph	Photon-like pulse ≈ 36 pC / 10 ns into A (sim-only)
J3/J4/J5/JMP/DNP	`exclude_from_sim`

PMT model is **intentionally simplified** (resistive bias + pulsed anode current).

Mech prep (agent or operator)

```
bash sim/spice/run_bringup_sims.sh
```

Expect exit 0 and `SPICE MECH OK` (not gate PASS).

Operator procedure — KiCad Simulator

Open `hardware/pmt-interface/bringup-supply.kicad_sch` in the schematic editor (`eeschema` or KiCad project → page 2).

0. Open the simulator

1. **Inspect → Simulator** (or the simulator toolbar icon).
2. Confirm sim sources: **Vhv** (left of HV), and near anode **Iph**, **Ra**, **Ca**.
3. Connectors J3/J4/J5 excluded from sim — expected.

1. Voltage probes — divider (DC / OP)

Analysis: Operating Point (DC).

Probe these **global** nets (voltage probe on the label):

Probe	Expect @ Vhv = −1000 V
K	−1000 V
D1	≈ −818.2 V
D2	≈ −681.8 V
D3	≈ −545.5 V
D4	≈ −409.1 V
D5	≈ −318.2 V
D6	≈ −227.3 V
D7	≈ −136.4 V
D8	≈ −45.5 V
A (idle)	≈ 0 V

Optional: current through **Vhv** ≈ **87 µA** (string ~41 µA + bleed ~45 µA).

Step 1 PASS: D1 and D8 within ~2% of table; voltages step monotonically toward GND down the string.

2. Stimulus — photon current pulse

Iph (sim-only): `PULSE(0 3.6m 50n 0.5n 0.5n 10n 1)`

Param	Meaning
0 → 3.6 mA	Peak conventional current **A → GND** (electrons into anode)
td = 50 ns	Pulse starts
tw = 10 ns	Width → charge ≈ 36 pC
per = 1 s	Single pulse in a short TRAN window

Analysis: Transient, e.g. `0` → `200`n, step ~`0.2`n.

Optional: probe **I(Iph)** — expect 3.6 mA pulse near 50–60 ns.

3. Anode response to stimulus

Keep TRAN. Probe **V(A)** (optional: current through **Ra**).

What	Expect
Before ~50 ns	V(A) ≈ 0
During/after pulse	V(A) goes **negative**
Min in ~40–120 ns	**A_MIN ≈ −2.5 V** (with Ca = 15 pF)
Shape	Brief negative spike, then recover toward 0

Step 3 PASS: anode clearly negative; amplitude volts-scale (not ~kV — that means Ca missing).

Alternate: export review

Instead of (or in addition to) the GUI: open `kicad-out/bringup-supply.cir` and the OP/TRAN console from the mech run; confirm the same numbers.

Checklist

#	Check	Mech	Operator
1	Export has Vhv, Iph, Ra, R10–R18 (no __ stubs)	[ ]	[ ]

#	Check	Mech	Operator
2	OP probes: D1 ≈ −818.18 V, D8 ≈ −45.45 V @ −1000 V	[ ]	[ ]
3	OP: |I(Vhv)| ≈ 87 µA (string + 22 M bleed)	[ ]	[ ]
4	TRAN: anode **negative** (A_MIN ≈ −2.5 V w/ Ca)	[ ]	[ ]
5	Operator agrees KiCad result is acceptable	—	[ ]

Sign-off

Field	Value
Date	
Operator	
KiCad path used	Simulator / export review
Result	**PASS** / FAIL

Blank Result = Gate 0c **Open**. Agent must not mark PASS.

After operator PASS

1. Unlock schematic redraw: R10A/B series, R11–R13 = 3.3M, R14–R17 = 2.2M, R18A‖R18B.
2. Re-run validate + Gate 0b spot-check on network refs.
3. Gate A build (`gate-a-build.md`).